Hydrogenic atoms

$$\hat{H}_0\psi = -\frac{\hbar^2}{2m}\vec{\nabla}^2\psi + V(r)\psi = E\psi$$

Laplacian $$\vec{\nabla}^2 = \frac{1}{r^2}\frac{\partial}{\partial r}\left(r^2\frac{\partial}{\partial r}\right) + \frac{1}{r^2}\underbrace{\left[\frac{1}{\sin\theta}\frac{\partial}{\partial\theta}\left(\sin\theta\frac{\partial}{\partial\theta}\right) + \frac{1}{\sin^2\theta}\frac{\partial^2}{\partial\varphi^2}\right]}_{-\hat{\ell}^2}$$

($\hat{\ell}$ = dimensionless angular momentum)

$$\frac{\hbar^2}{2m}\left\{-\frac{1}{r^2}\frac{\partial}{\partial r}\left(r^2\frac{\partial}{\partial r}\right) + \frac{\hat{\ell}^2}{r^2}\right\}\psi + V(r)\psi = E\psi$$

$$V(r) \to \frac{-Ze^2}{4\pi\varepsilon_0 r}$$

$$\psi(\vec{r}) = R_{n\ell}(r)\, Y_{\ell m}(\theta,\varphi)$$

spherical harmonics: $\hat{\ell}^2 Y_{\ell,m}(\theta,\varphi) = \ell(\ell+1)\, Y_{\ell m}(\theta,\varphi)$

$$\vec{L}^2 = \hbar^2\hat{\ell}^2$$

$$E_n^0 = -\frac{m}{2\hbar^2}\left(\frac{Ze^2}{4\pi\varepsilon_0}\right)^2\frac{1}{n^2} = -\frac{(Ze)^2}{2a_0 n^2}$$

$\ell = 0, 1 \ldots n-1.$

$$a_0 = \frac{4\pi\varepsilon_0\hbar^2}{me^2} = \text{Bohr radius}$$

$$m = \underbrace{-\ell, \ldots, \ell-1, \ell}_{2\ell+1}$$

Total degeneracy $= \underset{\text{spin}}{(2)} \cdot n^2$

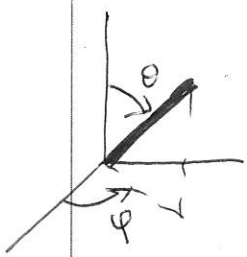
Radial dependence $R_{n\ell}(r) \sim e^{-r/\left(\frac{na_0}{Z}\right)}$

Magnetism is <u>absent</u> classically $\Rightarrow$ purely quantum effect.

Hydrogen atom

$$-\frac{\hbar^2}{2m}\nabla^2\psi + V(r)\psi = E\psi$$

$$\nabla^2 = \frac{1}{r^2}\frac{\partial}{\partial r}\left(r^2\frac{\partial}{\partial r}\right) + \frac{1}{r^2\sin\theta}\left(\frac{\partial}{\partial\theta}\sin\theta\frac{\partial}{\partial\theta}\right) + \frac{1}{r^2\sin^2\theta}\frac{\partial^2}{\partial\varphi^2}$$



$$\psi(r,\theta,\varphi) = R(r)\,Y(\theta,\varphi)$$

<u>Angular equation:</u> $\sin\theta\frac{\partial}{\partial\theta}\left(\sin\theta\frac{\partial Y}{\partial\theta}\right) + \frac{\partial^2 Y}{\partial\varphi^2} = -l(l+1)\sin^2\theta\, Y$

spherical harmonics $\hat{L}^2 = -\hbar^2\left[\frac{1}{\sin\theta}\frac{\partial}{\partial\theta}\left(\sin\theta\frac{\partial}{\partial\theta}\right) + \frac{1}{\sin^2\theta}\frac{\partial^2}{\partial\varphi^2}\right]$

$$Y_l^m(\theta,\varphi) = (\varepsilon)\sqrt{\frac{2l+1}{4\pi}\frac{(l-|m|)!}{(l+|m|)!}}\; e^{im\varphi}\, P_l^m(\cos\theta)$$

(associated Legendre function)

$\varepsilon = 1$ for $m \le 0$
$\varepsilon = (-1)^m$ $m > 0$

$$Y_0^0 = \frac{1}{\sqrt{4\pi}} \qquad \int_0^{2\pi}\int_0^{\pi} |Y_l^m(\theta,\varphi)|^2 \sin\theta\, d\theta\, d\varphi = 1$$

$$Y_1^0 = \sqrt{\frac{3}{4\pi}}\cos\theta = \sqrt{\frac{3}{4\pi}}\frac{z}{r}$$

$$Y_1^{\pm 1} = \mp\left(\frac{3}{8\pi}\right)^{1/2}\sin\theta\, e^{\pm i\varphi} = \mp\left(\frac{3}{8\pi}\right)^{1/2}\frac{x \pm iy}{r}$$

$$Y_2^0 = \left(\frac{5}{16\pi}\right)^{1/2}(3\cos^2\theta - 1)$$

$$Y_2^{\pm 1} = \mp\left(\frac{15}{8\pi}\right)^{1/2}\sin\theta\cos\theta\, e^{\pm i\varphi}$$

$$Y_2^{\pm 2} = \left(\frac{15}{32\pi}\right)^{1/2}\sin^2\theta\, e^{\pm i2\varphi}$$

Radial equation $\quad R(r) = \dfrac{u(r)}{r}$

$$-\frac{\hbar^2}{2m}\frac{d^2u}{dr^2} + \underbrace{\left[V(r) + \frac{\hbar^2}{2m}\frac{l(l+1)}{r^2}\right]}_{\text{effective potential}} u = E u$$

Hydrogen line $\quad V(r) = -\dfrac{Ze^2}{4\pi\varepsilon_0}\dfrac{1}{r}$

(does not depend on l) $\quad E_n = -\left[\dfrac{m}{2\hbar^2}\left(\dfrac{e^2}{4\pi\varepsilon_0}\right)^2\right]\dfrac{1}{n^2} = \dfrac{E_1}{n^2}$ $\quad n = 1, 2, 3 \ldots$ principal quantum number

$E_n = -\dfrac{\hbar^2}{2ma^2}\dfrac{1}{n^2}$ particle in a box of size a

Bohr radius $\quad a = \dfrac{4\pi\varepsilon_0\hbar^2}{me^2} = 0.53\cdot 10^{-10}\ m \quad (0.5\,\text{Å})$

(remember $\rho = r/an$) $\quad a = \dfrac{\hbar}{mc\alpha}$ $\quad \alpha = \dfrac{e^2}{4\pi\varepsilon_0\hbar c} \simeq \dfrac{1}{137}$ fine structure constant

$$R_{10} = 2a^{-3/2}e^{-r/a}$$

$$R_{20} = \frac{1}{\sqrt{2}}a^{-3/2}\left(1 - \frac{r}{2a}\right)e^{-r/2a}$$

$$R_{21} = \frac{1}{\sqrt{24}}a^{-3/2}\left(\frac{r}{a}\right)e^{-r/2a}$$

p.154 $\quad R_{30} = \dfrac{2}{\sqrt{27}}a^{-3/2}\left(1 - \dfrac{2r}{3a} + \dfrac{2}{27}\dfrac{r^2}{a^2}\right)e^{-r/3a}$

etc.

Observe that $R_{nl} \sim e^{-\frac{r}{na}}$: $n>1$ levels are closer to the nuclei while $l>0$ states are "pushed" away from the center

Direct exchange

- Idea: e-e interaction between electrons in the same shell can split (some of the) degeneracies!

$$\hat{H} = \hat{H}_0(\vec{r}_1) + \hat{H}_0(\vec{r}_2) + \frac{e^2}{4\pi\varepsilon_0 |\vec{r}_1 - \vec{r}_2|}$$

Orbital w.f. $H_0(\vec{r})\,\varphi_{a,b}(\vec{r}) = \varepsilon_{a,b}\,\varphi_{a,b}(\vec{r})$.

<u>orthogonal</u> $\int \varphi_a^*(r)\,\varphi_b(r)\,d\vec{r} = 0$

Two electrons $\Rightarrow$ 4-fold degenerate states with $\varepsilon = \varepsilon_a + \varepsilon_b$.

Proper basis $\rightarrow$ Slater determinants for many-electron w.f.

$$\Psi(\bar{r}_1, \bar{r}_2 \ldots \bar{r}_N) = \frac{1}{\sqrt{N}} \begin{vmatrix} \varphi_1(r_1) & \varphi_1(r_2) & \ldots & \varphi_1(r_N) \\ \varphi_2(r_1) & \varphi_2(r_2) & \ldots & \varphi_2(r_N) \\ \vdots & & & \\ \varphi_N(r_1) & \varphi_N(r_2) & \ldots & \varphi_N(r_N) \end{vmatrix}$$

state index

electron index

Remember to include spin w.f. $\chi(S) \rightarrow$ [denote α for $\uparrow$, β for $\downarrow$

$$\Psi_1 = \frac{1}{\sqrt{2}} \begin{vmatrix} \varphi_a(r_1)\alpha_1 & \varphi_\alpha(r_2)\alpha_2 \\ \varphi_b(r_1)\alpha_1 & \varphi_b(r_2)\alpha_2 \end{vmatrix} = \frac{1}{\sqrt{2}}\alpha_1\alpha_2\big(\varphi_a(r_1)\varphi_b(r_2) - \varphi_a(r_2)\varphi_b(r_1)\big)$$

Both e's have spin $\uparrow$ up

$$\psi_4 = \frac{1}{\sqrt{2}} \begin{vmatrix} \varphi_a(r_1)\beta_1 & \varphi_a(r_2)\beta_2 \\ \varphi_b(r_1)\beta_1 & \varphi_b(r_2)\beta_2 \end{vmatrix} = \frac{1}{\sqrt{2}} \beta_1\beta_2 \Big(\varphi_a(r_1)\varphi_b(r_2) - \varphi_a(r_2)\varphi_b(r_1)\Big)$$

$$\psi_2 = \frac{1}{\sqrt{2}} \begin{vmatrix} \varphi_a(r_1)\beta_1 & \varphi_a(r_2)\beta_2 \\ \varphi_b(r_1)\alpha_1 & \varphi_b(r_2)\alpha_2 \end{vmatrix} =$$

$$= \frac{1}{\sqrt{2}} \Big(\varphi_a(r_1)\varphi_b(r_2)\,\beta_1\alpha_2 - \varphi_b(r_1)\varphi_a(r_2)\,\alpha_1\beta_2\Big)$$

(No factorization of orbital and spin part!)

$$\psi_3 = \frac{1}{\sqrt{2}} \begin{vmatrix} \varphi_a(r_1)\alpha_1 & \varphi_a(r_2)\alpha_2 \\ \varphi_b(r_1)\beta_1 & \varphi_b(r_2)\beta_2 \end{vmatrix} =$$

$$= \frac{1}{\sqrt{2}} \Big(\varphi_a(r_1)\varphi_b(r_2)\,\alpha_1\beta_2 - \varphi_b(r_1)\varphi_a(r_2)\,\beta_1\alpha_2\Big)$$

Diagonalize $\hat{\mathcal{H}}$ in subspace $\psi_{1\ldots4}$ $\Rightarrow$ need matrix elements.

$$\langle\psi_1| \frac{e^2}{|r_1-r_2|} |\psi_1\rangle = \frac{1}{2}\int dr_1\, dr_2 \Big(\varphi_a^*(r_1)\varphi_b^*(r_2) - \varphi_a^*(r_2)\varphi_b^*(r_1)\Big) \cdot \frac{e^2}{|r_1-r_2|}\Big(\varphi_a(r_1)\varphi_b(r_2) - \varphi_a(r_2)\varphi_b(r_1)\Big)$$

$$= C_{ab} - J_{ab}$$

$$C_{ab} = e^2 \int dr_1 dr_2 \; \frac{|\varphi_a(r_1)|^2 \, |\varphi_b(r_2)|^2}{|r_1 - r_2|} = e^2 \int dr_1 dr_2 \; \frac{|\varphi_a(r_2)|^2 \, |\varphi_b(r_1)|^2}{|r_1 - r_2|}$$

$$J_{ab} = e^2 \int dr_1 dr_2 \; \frac{\varphi_a^*(r_1)\, \varphi_b(r_1)\; \varphi_b^*(r_2)\, \varphi_a(r_2)}{|r_1 - r_2|}$$

$$= e^2 \int dr_1 dr_2 \; \frac{\varphi_a^*(r_2)\, \varphi_b(r_2)\; \varphi_b^*(r_1)\, \varphi_a(r_1)}{|r_1 - r_2|}$$

(just interchange $\vec{r}_1 \leftrightarrow \vec{r}_2$)

$$\langle \Psi_2 | \frac{e^2}{|r_1 - r_2|} | \Psi_2 \rangle = C_{ab}$$

(J_{ab} absent because $\beta_1^* \alpha_2^* \alpha_1 \beta_2 = 0$!)

but $\langle \Psi_2 | \frac{e^2}{|r_1 - r_2|} | \Psi_3 \rangle = -J_{ab}$!

$$\langle \Psi_4 | \frac{e^2}{|r_1 - r_2|} | \Psi_4 \rangle = C_{ab} - J_{ab}$$

Matrix form

$$\Rightarrow \hat{\mathcal{H}} \rightarrow (\varepsilon_a + \varepsilon_b) \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} +$$

$$+ \begin{bmatrix} C_{ab} - J_{ab} & 0 & 0 & 0 \\ 0 & C_{ab} & -J_{ab} & 0 \\ 0 & -J_{ab} & C_{ab} & 0 \\ 0 & 0 & 0 & C_{ab} - J_{ab} \end{bmatrix}$$

$\Rightarrow$ $\psi_{1,4}$ are eigenstates of full $\mathcal{H}$, with $S^z_{tot} = \pm 1$

$$E_{1,4} = \varepsilon_a + \varepsilon_b + C_{ab} - J_{ab}$$

Diagonalize 2×2 subspace of $\psi_{2,3}$ states:

$$\det \begin{pmatrix} C_{ab} - E & -J_{ab} \\ -J_{ab} & C_{ab} - E \end{pmatrix} = (C_{ab} - E)^2 - J_{ab}^2 = 0$$

$$\Rightarrow E_\pm = \varepsilon_a + \varepsilon_b + C_{ab} \pm J_{ab}$$

$$\begin{pmatrix} \varepsilon_a + \varepsilon_b + C_{ab} & -J_{ab} \\ -J_{ab} & \varepsilon_a + \varepsilon_b + C_{ab} \end{pmatrix} \begin{pmatrix} f_1 \\ f_2 \end{pmatrix} = E_\pm \begin{pmatrix} f_1 \\ f_2 \end{pmatrix}$$

$(+)$ $$\left.\begin{aligned} -J f_2 &= J f_1 \\ -J f_1 &= J f_2 \end{aligned}\right\} \Rightarrow f_2 = -f_1 \Rightarrow f_1 = -f_2 = -\frac{1}{\sqrt{2}}$$

$\ominus$ $\quad -Jf_2 = -Jf_1$

$\quad -Jf_1 = -Jf_2 \quad \Rightarrow f_1 = f_2 \Rightarrow f_1 = f_2 = \frac{1}{\sqrt{2}}$

$\Rightarrow$ E_+ has (normalized) eigenstate $\psi_+ = \frac{1}{\sqrt{2}}(\psi_2 - \psi_3)$

E_- $\qquad \psi_- = \frac{1}{\sqrt{2}}(\psi_2 + \psi_3)$

But $E_- = E_{1,4} = E_t$ (triplet energy)

$\psi_1 \to |1,1\rangle$

$\psi_4 \to |1,-1\rangle$

$$\psi_- = \frac{1}{\sqrt{2}}(\psi_2 + \psi_3) = \frac{1}{2}\Big[\varphi_a(r_1)\varphi_b(r_2)(\beta_1\alpha_2 + \alpha_1\beta_2) - \varphi_b(r_1)\varphi_a(r_2)(\alpha_1\beta_2 + \beta_1\alpha_2)\Big]$$

$$= \frac{1}{\sqrt{2}}\Big(\varphi_a(r_1)\varphi_b(r_2) - \varphi_b(r_1)\varphi_a(r_2)\Big)\frac{1}{\sqrt{2}}(\alpha_1\beta_2 + \beta_1\alpha_2)$$

$\to |1,0\rangle$

Here $|S, M_z\rangle$ is the state of total spin S

z projection M_z

$E_+ = \varepsilon_a + \varepsilon_b + C_{ab} + J_{ab} = E_s$ singlet energy

$$\psi_+ = \frac{1}{\sqrt{2}}(\psi_2 - \psi_3) = \frac{1}{\sqrt{2}}\Big(\varphi_a(r_1)\varphi_b(r_2) + \varphi_b(r_1)\varphi_a(r_2)\Big)\cdot$$

$$\cdot\frac{1}{\sqrt{2}}(\alpha_1\beta_2 - \beta_1\alpha_2) \to |0,0\rangle$$

Useful to notice that $\Psi_\pm$ are linear superpositions of Slater determinants.

$$E_t - E_S = -2J_{ab}$$

Can in fact proof that here $J_{ab} > 0$:

Write $J_{ab} = \int dr_1 dr_2 \; W^*(r_1) f(r_1 - r_2) w(r_2)$

where $W(r) = \varphi_a(r) \varphi_b^*(r)$

$$\int dr_2 \; f(r_1 - r_2) \; w(r_2) = \int d^3\vec{k} \; e^{i\vec{k}\vec{r}} w(k) f(k)$$

$$\Rightarrow J_{ab} = \int w^*(-k) f(k) w(k) \, d^3\vec{k} =$$

$$= \int |w(k)|^2 f(k) \, d\vec{k} > 0$$

$$f(k) = \frac{4\pi e^2}{k^2} > 0$$

Thus: 1) triplet and singlet have different energy due to Coulomb interaction

2) in this case triplet has lower energy (for orbitals centered at different points $J_{ab} < 0$ resulting in singlet being <u>lower</u> in energy!)

$$\vec{S}_{tot} = \vec{S}_1 + \vec{S}_2$$

In our 4x4 space we can write

$$\hat{H} = A + B\,\vec{S}_{tot}^{\,2}$$

such that $\hat{H}\,|\text{triplet}\rangle = E_t\,|\text{triplet}\rangle$

$\hat{H}\,|\text{singlet}\rangle = E_s\,|\text{singlet}\rangle$

$$\Rightarrow \quad A = E_s, \quad A + B\,s(s+1) = A + 2B = E_t$$

$$B = \frac{1}{2}\left(E_t - E_s\right) = -J_{ab}$$

$$\hat{H} = E_s - J_{ab}\underbrace{\left(\vec{S}_1 + \vec{S}_2\right)^2}_{\underbrace{\vec{S}_1^{\,2} + \vec{S}_2^{\,2}}_{2\cdot\frac{1}{2}\left(\frac{1}{2}+1\right) = \frac{3}{2}} + 2\,\vec{S}_1\,\vec{S}_2}$$

$$\hat{H} = E_s - \frac{3}{2}J_{ab} - 2J_{ab}\,\vec{S}_1 \cdot \vec{S}_2$$

Heisenberg exchange between $\vec{S}_{1,2}$
due to Pauli principle
and Coulomb interaction only!